LESSON PLAN

PROPELLER THEORY (7)

Class	Air Agniveer Apprentice (Engineering)	**Lesson Plan No.**	05
Total No. of Lesson Plans	8	**Time Allotted for the Session**	80 min

PART A: GUIDELINES / GENERAL INSTRUCTIONS FOR INSTRUCTORS

1. Topic	Effect of propeller rotation on aircraft and methods used to counteract it
2. Total No. of Sessions for this Topic	01 / 01
3. Sequence of this Session	01 / 01
4. Syllabus Extract	Effect of propeller rotation on aircraft and methods used to counteract it
5. Specific Objectives	7.1.1.1 Trainee will be able to describe the various types of propellers, principles of operation and pitch change mechanisms.

6. List of Training Aids

Ser	Description	Remarks
(a)	BR/Text Book/Docket	Refer to the master syllabus reference list
(b)	Wall Charts	Nil
(c)	Films/Duration	Nil
(d)	PPT	Objective: teach the syllabus extract for this lesson
(e)	Models	Nil
(f)	CBT	Nil

7. Practical in Lab/Equipment Rooms/Simulators	Review of PPT diagrams; Instructor supervision required for safety during discussion of aircraft movement.
8. Existing Knowledge of the Class	As per course prerequisites

PART B: DEVELOPMENT OF LESSON

9. Checking of Previous Knowledge / Introduction	Time: 10 min

(a) What is the primary function of a propeller?
(b) How does a propeller generate thrust?
(c) Why is propeller rotation direction important?

10. Development of Main Lesson

Sub	Content	Method	Trg Aids	Time
(a)	• Definition of P-factor • Description of Gyroscopic Precession • Explanation of Torque Reaction • Identification of Spiral Slipstream • Overview of Propeller Efficiency	L & Ds	PPT, IB	15 min
(b)	• Principle of Torque Counteraction • Function of Tail Rudder • Role of Engine Mounting • Effect of Propeller Rotation Direction • Impact on Aircraft Handling	L & Ds	PPT, PDF	20 min
(c)	• Methods to Counteract P-factor • Techniques for Gyroscopic Precession • Strategies for Spiral Slipstream • Importance of Pilot Input • Summary of Countermeasures	Practical	IB, PDF	15 min

11. Summary and Evaluation

Method	Trg Aids	Time
Q & A	IB	15 min

(a) Explain how torque reaction affects aircraft yaw.
(b) Describe the relationship between propeller rotation and P-factor.
(c) List three methods used to counteract gyroscopic precession.

12. Home Assignment and Follow Up	Prepare a brief note on how the principles discussed in this lesson apply to the constructional details of the Dornier A/C propeller covered in LP 02.
Time	5 min

PART C: PREPARATION FOR TOMORROW (PFT) DIRECTIVES FOR TRAINEES

Ser	Syllabus for PFT	References	Related LPs
1	Constructional details about Dornier A/C propeller, construction of blades and pitch change mechanism and spinner	Refer to the master syllabus reference list	LP 06

PART D: PERSONAL EXPERIENCES (PE) OF INSTRUCTOR

Ser	Item	Remarks
(a)	Adequacy of time	
(b)	Suitability of training material/aids and additional aids required	
(c)	Doubts/intelligent questions raised by trainees	
(d)	Difficulties faced by instructor	
(e)	Other experience	

Instructor	**HoF (E/L)**
Signature: _____	Signature: _____

PART E: REVIEW SHEET

Date	Instructor's Comment	HOF (E/L)	Oi/C TDEC Remarks

Prepared by / Approved by

Prepared by	Approved by
Name, Rank, P No.	Name, Rank, P No.
Nov 26	Nov 26

Queries for Instructor

1. Confirm the master syllabus reference book/BR for this subject.
2. Confirm the class existing knowledge and prerequisites.
3. Confirm the session duration and sequence; 80 min and 01/01 were assumed.
4. Prepared-by/approved-by names, ranks, P numbers and dates must be filled before approval.
5. Confirm any training aids available beyond L & Ds, IB, PDF and PPT.

Timing Check

Introduction 10 + Main 50 + Summary 15 + Home 5 = 80 min